

Responsibility in Automated Software Testing

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Ethical Standing on Therac-25 Incident

The Therac-25 incident was an important event that occurred to alter the way we think about making moral decisions related to software development. When developing and testing software for the purpose of saving lives, taking extra precautions to make sure that a systems software works as intended is very crucial. To dictate the appropriate amount of testing needed, developers need to consider both the potential vulnerabilities, risks and budget involved. An expertise plan should include ways to minimize the risks and budget, but to never to fully value the budget over the safety of everyone involved. There have been many cases during recent times showing how malfunctions of hardware due to software errors can happen effortlessly, leading to people experiencing undesirable circumstances. Consistently evaluating systems needs to be required by any professional business to lessen the dilemmas that can arise from the systems software. It is morally correct to value the lives of others than the cost impact, as nobody would want others to devalue the cost constraints over the value of life God has granted everyone.

A system that is made with the vision to help people but is considered to not be reliable should not be something to put up for sale. Practicing this makes it seem like a situation of overpromising whilst underdelivering which can cause major predicaments to those involved. Although the cheaper and unreliable systems are more accessible to those in need, the risks involved with it make the system not worth selling and instead should put more resources into making it safer. Luke 6:31 states, “Do to others as you would have them do to you” (New International Version). Other alternative remedies I believe would work include petitioning to receive extra funding from resources like the government or a charity to help with budget. My thoughts thoroughly align with the ACM principles. The Association for Computing Machinery (2018) states, “Contribute to society and to human well-being, acknowledging that all people are

stakeholders in computing” (p. 1). To give aid towards society while acknowledging that everyone involved is a stakeholder assembles a secure ethical foundation for many to consider. If I were able to sell this system, I would not sell it from an ethical standpoint because of the many potential dilemmas involved with the system.

It is common in business that engineers require certifications to be deemed eligible and receive the skills necessary to develop a reliable product. This is not the case for software engineers, so this brings up a popular discussion on whether there should be a similar certification process for them. I believe this proposal will drastically increase the reliability of software engineers on top of also minimizing risks of implementing faulty and non-reliable systems that could jeopardize individuals. There are critical concerns from many people that software engineers do not have to go through a process to prove their credibility. Failure in inciting a proper process for software engineers to undergo can lead to trust complications between people. 1 Corinthians 8:9 states “Be careful, however, that the exercise of your rights does not become a stumbling block to the weak” (New International Version). Implementing a process with features like this establishes a sense that they will do their best to cause unintended interference with systems, and that it should be thought of carefully and implemented in ways that suit the workplace environment as there can be certain circumstances that cannot be accounted for. The Institute of Electrical and Electronics Engineers (2025) have a critical ethical rule, stating, “To seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors” (p. 5). To preserve integrity and morality is why I believe incorporating this process of a certification is needed and is one of many ways the Therac-25 incident could have been prevented.

References

Association for Computing Machinery. (2018). *The code affirms an obligation of computing professionals to use their skills for the benefit of society*. Code of Ethics.

<https://ethics.acm.org/code-of-ethics>

Institute of Electrical and Electronics Engineers. (2025). IEEE code of ethics

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